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Drying device and printing apparatus

Abstract

A drying device includes a support unit, a heating unit, and a control unit. The heating unit includes a duct, which has a blowing outlet, a suction inlet, and an air port and includes an air flow path, a heating section, a blower, and a first adjustment unit capable of adjusting an amount of air flowing into the air flow path from an outside of the duct through the air port. The control unit is capable of executing a first mode in which the first adjustment unit is controlled to adjust the amount of air in a state in which the heating section performs heating and a second mode in which the first adjustment unit is controlled to cause the amount of air to be more than the amount of air in the first mode in a state in which heating by the heating section is stopped.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-021889, filed Feb. 16, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a drying device and a printing apparatus.

2. Related Art

(3) A drying device described in JP-A-2019-107822 includes a heating unit that heats a medium

supported on a support surface. The heating unit includes a heating section that heats a medium, a housing being an example of a duct, an air flow path, and a blower for sending air through the air flow path. A suction inlet and a blowing outlet are formed in the housing. The suction inlet and the blowing outlet are opened toward the support surface, and communicate with the air flow path. The air flowing through the air flow path is discharged through the blowing outlet to a region between the heating unit and the support surface. Part of the air in the region between the heating unit and the support surface flows through the suction inlet into the air flow path. In this manner, the air circulates between the region between the heating unit and the support surface, and the air flow path.

(4) In the drying device, even when heating of the heating section is stopped after completion of drying of the medium, a temperature of the air in the region between the heating unit and the support surface is not likely to be lowered. Therefore, a thermal damage may be caused to the medium supported on the support surface.

SUMMARY

(5) In order to solve the above-mentioned problem, a drying device includes a support unit configured to support, on a support surface, a medium transported after printing, a heating unit facing the support surface and being configured to heat the medium supported on the support surface after printing, and a control unit configured to control the heating unit, wherein the heating unit includes a duct having a blowing outlet through which air is blown out toward the support surface, a suction inlet through which the air blown out through the blowing outlet toward the support surface is sucked, and an air port being different from the blowing outlet and the suction inlet, and including an air flow path through which the air is flowable, a heating section configured to heat at least one of the air in the air flow path and the medium supported on the support surface, a blower configured to cause the air in the air flow path to flow toward the blowing outlet, and a first adjustment unit configured to adjust an amount of air flowing into the air flow path from an outside of the duct through the air port, the duct has a facing region facing the support surface and a non-facing region not facing the support surface, the air port is provided in the non-facing region, and the control unit is configured to execute a first mode in which the first adjustment unit is controlled to adjust the amount of air in a state in which the heating section performs heating and a second mode in which the first adjustment unit is controlled to cause the amount of air to be more than the amount of air in the first mode in a state in which heating by the heating section is stopped.

(6) In order to solve the above-mentioned problem, a printing apparatus includes a transport unit configured to transport a medium, a printing unit configured to perform printing on the medium transported by the transport unit, a support unit configured to support, on a support surface, the medium after printing performed by the printing unit, a heating unit facing the support surface and being configured to heat the medium supported on the support surface after printing, and a control unit configured to control the heating unit, wherein the heating unit includes a duct having a blowing outlet through which air is blown out toward the support surface, a suction inlet through which the air blown out through the blowing outlet toward the support surface is sucked, and an air port being different from the blowing outlet and the suction inlet, and including an air flow path through which the air is flowable, a heating section configured to heat at least one of the air in the air flow path and the medium supported on the support surface, a blower configured to cause the air in the air flow path to flow toward the blowing outlet, and a first adjustment unit configured to adjust an amount of air flowing into the air flow path from an outside of the duct through the air port, the duct has a facing region facing the support surface and a non-facing region not facing the support surface, the air port is provided in the non-facing region, and the control unit is configured to execute a first mode in which the first adjustment unit is controlled to adjust the amount of air in a state in which the heating section performs heating and a second mode in which the first adjustment unit is controlled to cause the amount of air to be more than the amount of air in the first mode in a state in which heating by the heating section is stopped.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic view illustrating a printing apparatus according to an exemplary embodiment.
- (2) FIG. 2 is a cross-sectional view illustrating a drying device.
- (3) FIG. 3 is a block diagram illustrating an electrical configuration of the printing apparatus and the drying device.
- (4) FIG. 4 is a flowchart showing a first mode execution routine.
- (5) FIG. 5 is a flowchart showing a second mode execution routine.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(6) A drying device and a printing apparatus according to an exemplary embodiment are described below with reference to the drawings. The printing apparatus is, for example, an ink-jet type printer that performs printing by ejecting ink being an example of a liquid onto a medium such as paper, fabric, vinyl, a plastic component, and a metal component.

(7) In the drawings, the direction of gravity is indicated by a Z axis while assuming that the printing apparatus is placed on a horizontal plane, and directions along the horizontal plane are indicated by an X axis and a Y axis. The X axis, the Y axis, and the Z axis are orthogonal to one another. A direction parallel to the X axis is also referred to as a width direction X. A direction parallel to the Y axis is also referred to as a depth direction Y. A direction parallel to the Z axis is also referred to as a vertical direction Z.

(8) Printing Apparatus

(9) As illustrated in FIG. 1, a printing apparatus **10** may include a housing **13**. The printing apparatus **10** includes a printing unit **12** that performs printing on a medium M. The printing unit **12** may be positioned inside the housing **13**. The width direction X is also a width direction of the medium M. The printing unit **12** may include a carriage **12a** that moves in a scanning direction SD and a guide shaft **12b** that has a rod-like shape and extends in the width direction X. The scanning direction SD may be a direction parallel to the X axis. The carriage **12a** is supported by the guide shaft **12b**. The carriage **12a** is reciprocable along the guide shaft **12b** by driving a motor, which is not illustrated. The carriage **12a** and the guide shaft **12b** are away above from the medium M transported by a transport unit **22**.

(10) The printing unit **12** may include a recording head **12c** that performs recording on the medium M. The recording head **12c** has a nozzle surface **12d** in which a nozzle **12h** is opened. The recording head **12c** is mounted on the carriage **12a**. The recording head **12c** moves together with the carriage **12a**, and thus performs scanning in the scanning direction SD. The recording head **12c** may perform recording on the medium M by ejecting ink being an example of a liquid from the nozzle **12h** onto the medium M. The recording head **12c** in the present exemplary embodiment is a serial type that performs printing while moving in the width direction X of the medium M. The recording head **12c** may be configured as a line type that is provided in the width direction X of the medium M.

(11) The printing apparatus **10** may include a printing support **11**. The printing support **11** is positioned inside the housing **13**. The printing support **11** is positioned below the recording head **12c**. The medium M is transported in an upstream transport direction D1 along the upper surface of the printing support **11**. The upstream transport direction D1 is a direction along the upper surface of the printing support **11** and the length direction of the medium M.

(12) Feeding Unit

(13) The printing apparatus **10** may include a feeding unit **14**. The feeding unit **14** feeds the medium M. The feeding unit **14** includes a support shaft **14a** and a feeding motor **14b**. The support shaft **14a** rotatably supports a roll body R. The roll body R is the medium M wound in a roll shape.

The feeding motor **14b** is a power source that rotates the support shaft **14a**. The feeding motor **14b** drives the support shaft **14a** to rotate, and thus the feeding unit **14** feeds out the elongated medium M from the roll body R. The feeding unit **14** feeds out the medium M from the roll body R to the inside of the housing **13**.

(14) Transport Unit

(15) The printing apparatus **10** includes the transport unit **22** that transports the medium M. The printing unit **12** performs printing on the medium M transported by the transport unit **22**. In other words, the printing unit **12** performs printing on the medium M transported by the transport unit **22**.

(16) The transport unit **22** may include a transport motor, which is not illustrated, and a transport roller pair **24**. In the drawing, the transport roller pair **24** being one pair is illustrated. Alternatively, the transport unit **22** may include a plurality of transport roller pairs **24**. The transport roller pair **24** may rotate in a state of sandwiching the medium M to transport the medium M. The transport motor is intermittently driven, and thus the transport unit **22** in the present exemplary embodiment intermittently transports the medium M. A transport operation performed by the transport unit **22** may be performed alternately with a printing operation performed by the printing unit **12**.

(17) Winding Unit

(18) The printing apparatus **10** may include a winding unit **15**. The winding unit **15** winds, into a roll shape, the medium M subjected to printing by the printing unit **12**. The winding unit **15** includes a winding shaft **15a** and a winding motor **15b**. The winding shaft **15a** rotatably supports the roll body R. The roll body R at the winding unit **15** is the medium M after printing, which is wound in a roll shape. The winding motor **15b** is a driving source of the winding shaft **15a**. The winding motor **15b** drives the winding shaft **15a** to rotate, and thus the winding unit **15** winds the medium M. The winding motor **15b** may be driven at timing later than driving of the feeding motor **14b**, and the driving thereof may be stopped at a timing later than stoppage of the driving of the feeding motor **14b**.

(19) Drying Device

(20) The printing apparatus **10** includes a drying device **20**. The drying device **20** includes a support unit **21**. In other words, it can also be understood that the printing apparatus **10** includes the support unit **21**. The support unit **21** may be positioned outside the housing **13**. The support unit **21** has a support surface **21a**. The support unit **21** supports, on the support surface **21a**, the medium M after printing, which is subjected to printing performed by the printing unit **12**. The support surface **21a** may extend between the housing **13** and the winding unit **15**. The dimension of the support surface **21a** in the width direction X may be larger than the dimension of the medium M in the width direction X.

(21) The medium M is transported in a transport direction D2 along the support surface **21a**. The transport direction D2 is a direction along the support surface **21a** and the length direction of the medium M. The support surface **21a** may extend so as to be displaced in the depth direction Y and inclined downward in the vertical direction Z as approaching downstream in the transport direction D2. In this case, the medium M transported along the support surface **21a** is displaced in the depth direction Y and transported downward in the vertical direction Z as approaching downstream in the transport direction D2.

(22) Heating Unit

(23) As illustrated in FIG. 2, the drying device **20** includes a heating unit **25**. In other words, it can also be understood that the printing apparatus **10** includes the heating unit **25**. The heating unit **25** faces the support surface **21a**. The heating unit **25** heats the medium M after printing, which is supported on the support surface **21a**. The heating unit **25** includes a duct **31**, a heating section **32**, a blower **33**, and a first adjustment unit **34**. The heating unit **25** may have a partition wall **42**. The heating unit **25** may include a second adjustment unit **52**.

(24) Duct

(25) The duct **31** may have an outer wall **41**. A space is defined and formed inside the duct **31** with

the outer wall **41**. The partition wall **42** may be positioned inside the duct **31**. A part of the outer wall **41** and the partition wall **42** form an air flow path **43** through which air can flow. The air flow path **43** is formed inside the duct **31**. In other words, the duct **31** includes the air flow path **43**.

(26) The outer wall **41** of the duct **31** may have a first outer wall **41a** and a second outer wall **41b**. The first outer wall **41a** may have a flat plate shape. The first outer wall **41a** may cause the air to be communicable. For example, the shape of the first outer wall **41a** may be a mesh-like shape or a shape in which a plurality of through holes are formed. The second outer wall **41b** is positioned to surround the first outer wall **41a**.

(27) The first outer wall **41a** has a first outer surface **41c**. The first outer surface **41c** is an outer surface of the first outer wall **41a**. The first outer surface **41c** is a surface facing the support surface **21a**. The first outer surface **41c** may be parallel to the support surface **21a**. The first outer surface **41c** and the support surface **21a** are positioned away from each other. A direction orthogonal to the support surface **21a** is referred to as an orthogonal direction **D3**. The orthogonal direction **D3** may be a direction orthogonal to the first outer surface **41c**. The first outer surface **41c** may face the support surface **21a** in the orthogonal direction **D3**.

(28) The partition wall **42** contains a heat insulating material **42a**. Specifically, the partition wall **42** may be formed of the heat insulating material **42a** and an inner wall **42b**. The heat insulating material **42a** is formed of a material having a heat insulating effect higher than the inner wall **42b**. The inner wall **42b** is formed of the same material as the outer wall **41**. The heat insulating material **42a** and the inner wall **42b** may be in contact with each other. The inner wall **42b** may be positioned between the heat insulating material **42a** and the second outer wall **41b**.

(29) The end of the partition wall **42** may be positioned between the end of the first outer wall **41a** and the end of the second outer wall **41b**. The end of the partition wall **42** may be fixed to the first outer wall **41a**. The second outer wall **41b** and the partition wall **42** may form the air flow path **43**. The partition wall **42** and the first outer wall **41a** may define and form an internal space **S** inside the duct **31**. The internal space **S** may be adjacent to the air flow path **43** through intermediation of the partition wall **42**.

(30) The duct **31** has a facing region **44** that faces the support surface **21a** and a non-facing region **45** that does not face the support surface **21a**. The facing region **44** in the present exemplary embodiment is formed of the entire first outer wall **41a** and the end of the second outer wall **41b**. The non-facing region **45** in the present exemplary embodiment is formed of a part of the second outer wall **41b**, which does not form the facing region **44**. The facing region **44** is away from the support surface **21a** in the orthogonal direction **D3**. A region between the facing region **44** and the support surface **21a** is referred to as a region **46**. The region **46** is a region between the heating unit **25** and the support surface **21a**.

(31) Suction Inlet

(32) A suction inlet **47** is provided in the duct **31**. The suction inlet **47** communicates with the air flow path **43**. In other words, the air flow path **43** has the suction inlet **47**. The suction inlet **47** is a through hole formed in the outer wall **41**. The inside and the outside of the duct **31** are communicable through the suction inlet **47**.

(33) The suction inlet **47** may be provided in the facing region **44**. The suction inlet **47** may be provided to the second outer wall **41b** at a position below the first outer wall **41a**. The suction inlet **47** may communicate with the end of the air flow path **43**. Part of the air in the region **46** can flow through the suction inlet **47** into the air flow path **43**.

(34) Blowing Outlet

(35) A blowing outlet **48** through which the air is blown out toward the support surface **21a** is formed in the duct **31**. The blowing outlet **48** communicates with the air flow path **43**. In other words, the air flow path **43** has the blowing outlet **48**. The blowing outlet **48** is a through hole formed in the outer wall **41**. The inside and the outside of the duct **31** are communicable through the blowing outlet **48**.

(36) The blowing outlet **48** is provided in the facing region **44**. The blowing outlet **48** may be provided to the second outer wall **41b** at a position above the first outer wall **41a**. The blowing outlet **48** may communicate with an end of both the ends of the air flow path **43** with which the suction inlet **47** does not communicate. The air flowing through the suction inlet **47** into the air flow path **43** flows toward the blowing outlet **48**. In other words, the blowing outlet **48** communicates with downstream of the suction inlet **47** in a flowing direction in which the air flows through the air flow path **43**. The suction inlet **47** is formed so as to such the air blown out through the blowing outlet **48** onto the support surface **21a** therethrough.

(37) Air Port

(38) An air port **49** that is different from the blowing outlet **48** and the suction inlet **47** is formed in the duct **31**. The air port **49** communicates with the air flow path **43**. In other words, the air flow path **43** has the air port **49**. The air port **49** is a through hole formed in the outer wall **41**. The inside and the outside of the duct **31** are communicable through the air port **49**.

(39) The air port **49** is provided in the non-facing region **45**. The air port **49** may be provided to the second outer wall **41b** at a position away from the suction inlet **47** and the blowing outlet **48**. The air port **49** communicates between the suction inlet **47** and the blowing outlet **48** in the air flow path **43**.

(40) A removal member **51** may be provided inside the air port **49**. In other words, the drying device **20** may include the removal member **51**. For example, the removal member **51** may have a mesh-like shape. For example, heat resistant plastic or a filter may be adopted as the removal member **51**. The removal member **51** removes a foreign matter from the air flowing through the air port **49**.

(41) Heating Section

(42) The heating section **32** heats the medium **M** supported on the support surface **21a**. The heating section **32** heats the surface of the medium **M** supported on the support surface **21a**, and thus causes the printing unit **12** to evaporate moisture of the liquid adhering to the medium **M**. With this, the medium **M** to which the liquid adheres is dried.

(43) The heating section **32** may be arranged in the internal space **S** of the duct **31**. In other words, the partition wall **42** provides partition between the air flow path **43** and the heating section **32** inside the duct **31**. The heating section **32** may heat the medium **M** supported on the support surface **21a**, via the first outer wall **41a**. The heating section **32** is aligned with a part of the air flow path **43** through intermediation of the partition wall **42** in the orthogonal direction **D3**. A portion of the air flow path **43**, which is aligned with the heating section **32** through intermediation of the partition wall **42** in the orthogonal direction **D3**, is referred to as a parallel installation portion **43a**. Note that the air port **49** is in communication with the air flow path downstream of the parallel installation portion **43a** in the flowing direction in which the air flows through the air flow path **43**.

(44) The heating section **32** may include a heat generating element **32a** and a reflecting plate **32b**. Examples of the heat generating element **32a** include a heater pipe, a halogen lamp, and the like. The heat generating element **32a** and the reflecting plate **32b** extend in the width direction **X**. The heating section **32** may include a plurality of heat generating elements **32a** and a plurality of reflecting plates **32b**. The heat generating element **32a** is away from the first outer wall **41a** in the orthogonal direction **D3**. The reflecting plate **32b** reflects heat radiated from the heat generating element **32a**. The reflecting plate **32b** is positioned between the heat generating element **32a** and the partition wall **42** in the orthogonal direction **D3**. The reflecting plate **32b** surrounds the back side of the heat generating element **32a**. The heat radiated from the heat generating element **32a** to the back side is reflected by the reflecting plate **32b** toward the first outer wall **41a**. With this, the heat is radiated from the first outer wall **41a** toward the support surface **21a**.

(45) Blower

(46) The blower **33** causes the air in the air flow path **43** to flow toward the blowing outlet **48**. The blower **33** may be arranged in the middle of the air flow path **43**. The position of the blower **33** in

the air flow path **43** may be downstream of the communication position of the air port **49** in the air flow path **43**. The blower **33** may include a fan **33a**. The blower **33** may rotate the fan **33a** to generate an air flow flowing toward the blowing outlet **48**. Along the air flow generated by the blower **33**, the air flows through the air flow path **43** from the suction inlet **47** to the blowing outlet **48**, and is also blown out through the blowing outlet **48**. In addition to heating of the medium M by the heating section **32**, blowing of the air onto the medium M by the blower **33** is performed. With this, the heating unit **25** performs drying of the medium M.

(47) First Adjustment Unit

(48) The first adjustment unit **34** is capable of adjusting an amount of air flowing into the air flow path **43** from the outside of the duct **31** through the air port **49**. Hereinafter, the amount of air flowing into the air flow path **43** from the outside of the duct **31** through the air port **49** is also referred to as an amount of air A. The first adjustment unit **34** may be positioned in the air flow path **43**. The first adjustment unit **34** may include a first main body portion **34a** in which an air hole **34h** is formed and a first closing member **34b** capable of closing the air hole **34h**.

(49) The first main body portion **34a** may be positioned along the inner surface of the second outer wall **41b** from the inside of the duct **31** so as to close the air port **49**. The air hole **34h** is a through hole passing through the first main body portion **34a**. The air port **49** and the air flow path **43** communicate with each other through the air hole **34h**. A plurality of air holes **34h** may be provided in the first main body portion **34a**.

(50) The first closing member **34b** can be displaced by a motor, which is not illustrated.

Displacement of the first closing member **34b** enables switching between a state in which the first closing member **34b** closes the air hole **34h** and a state in which the air hole **34h** is opened.

(51) The number of air holes **34h** to be opened may be adjusted by adjusting the position of the first closing member **34b**. It can be said that, as the number of air holes **34** closed by the first closing member **34b** is greater, the opening degree of the air port **49** of the duct **31** is smaller. It can be said that, when the first closing member **34b** closes all the air holes **34h** in the first adjustment unit **34**, the air port **49** is in a fully closed state. It can be said that, as the number of opened air holes **34** is greater, the opening degree of the air port **49** of the duct **31** is larger. It can be said that, when all the air holes **34h** in the first adjustment unit **34** are opened, the air port **49** is in a fully opened state.

(52) Second Adjustment Unit

(53) The second adjustment unit **52** is capable of adjusting the opening degree of the suction inlet **47**. The second adjustment unit **52** may be positioned in the air flow path **43**. The second adjustment unit **52** may include a second main body portion **52a** in which a suction hole **52h** is formed and a second closing member **52b** capable of closing the suction hole **52h**. The second main body portion **52a** may be positioned inside the duct **31** so as to close the suction inlet **47**. The suction hole **52h** is a through hole passing through the second main body portion **52a**. The region **46** and the air flow path **43** communicate with each other through the suction hole **52h**. A plurality of suction holes **52h** may be formed in the second main body portion **52a**. The second closing member **52b** may be positioned in the air flow path **43**. The second closing member **52b** can be displaced by a motor, which is not illustrated. Displacement of the second closing member **52b** enables switching between a state in which the second closing member **52b** closes the suction hole **52h** and a state in which the suction hole **52h** is opened.

(54) The number of suction holes **52h** to be opened may be adjusted by adjusting the position of the second closing member **52b**. It can be said that, as the number of suction holes **52h** closed by the second closing member **52b** is greater, the opening degree of the suction inlet **47** of the duct **31** is smaller. It can be said that, when the second closing member **52b** closes all the suction holes **52h** in the second adjustment unit **52**, the suction inlet **47** is in a fully closed state. It can be said that, as the number of suction holes **52h** is greater, the opening degree of the suction inlet **47** of the duct **31** is larger. It can be said that, when all the suction holes **52h** in the second adjustment unit **52** are opened, the suction inlet **47** is in a fully opened state.

(55) Temperature Detection Unit and Humidity Detection Unit

(56) As illustrated in FIG. 3, the heating unit **25** may include a temperature detection unit **53**. The temperature detection unit **53** detects an outside air temperature T . The heating unit **25** may include a humidity detection unit **54**. The humidity detection unit **54** detects humidity H in the region **46** between the heating unit **25** and the support surface **21a**. The humidity detection unit **54** may detect outside air humidity $H1$ being humidity of the outside air, in addition to the humidity H in the region **46**.

(57) Control Unit

(58) The printing apparatus **10** includes a control unit **35**. The control unit **35** controls various operations executed in the printing apparatus **10**. The control unit **35** controls the heating unit **25**. It can also be understood that the drying device **20** includes the control unit **35**.

(59) The control unit **35** may be configured as a circuit including α : one or more processors that perform various processes according to a computer program, β : one or more dedicated hardware circuits that perform at least some of the various processes, or γ : a combination thereof. The hardware circuit is, for example, an application-specific integrated circuit. The processor includes a CPU and a memory such as RAM and ROM, and the memory stores a program code or a command configured to cause the CPU to perform the process. The memory, that is, a computer readable medium includes all kinds of readable media accessible by a general purpose or dedicated computer.

(60) The control unit **35** may be electrically coupled to the transport unit **22**. When a printing condition is satisfied, the control unit **35** may drive the transport motor of the transport unit **22**. The printing condition may be satisfied when a printing instruction is input in response to an operation of the operation unit, which is not illustrated. The printing condition may be satisfied when a printing instruction is input from a terminal device, which is not illustrated. When the transport motor is driven, the transport unit **22** transports the medium M .

(61) The control unit **35** causes the transport unit **22** to perform intermittent transport and thus to transport the medium M . Intermittent transport is a switching operation between transport of the medium M by the transport unit **22** and stoppage of transport of the medium M in a predetermined cycle. Further, when transport of the medium M is stopped during intermittent transport, the control unit **35** causes the printing unit **12** to perform printing on the medium M .

(62) The control unit **35** may be electrically coupled to the heating section **32**. The control unit **35** may perform switching heat generation and stoppage of heat generation of the heat generating element **32a**. The heat generating element **32a** is heated, and thus the control unit **35** causes the heating section **32** to perform heating. Heat generation of the heat generating element **32a** is stopped, and thus the control unit **35** stops heating by the heating section **32**.

(63) The control unit **35** may be electrically coupled to the blower **33**. The control unit **35** may perform switching between driving and stoppage of driving of the blower **33**. When the blower **33** is driven, the fan **33a** of the blower **33** rotates, and thus the air flow flowing toward the blowing outlet **48** is generated. The control unit **35** may control an amount of the air in the air flow path **43**, which flows from the blower **33** toward the blowing outlet **48**, by adjusting a rotation speed of the fan **33a** of the blower **33**. The amount of the air in the air flow path **43**, which flows from the blower **33** toward the blowing outlet **48** is increased as the rotation speed of the fan **33a** is higher.

(64) The control unit **35** may be electrically coupled to the first adjustment unit **34**. The control unit **35** may adjust the position of the first closing member **34b** of the first adjustment unit **34**. In the present exemplary embodiment, the control unit **35** adjusts the number of opened air holes **34h** by adjusting the position of the first closing member **34b**. In other words, the control unit **35** is capable of controlling the opening degree of the air port **49** by controlling the first adjustment unit **34**.

(65) The control unit **35** may be electrically coupled to the second adjustment unit **52**. The control unit **35** may adjust the position of the second closing member **52b** of the second adjustment unit **52**. In the present exemplary embodiment, the control unit **35** adjusts the number of the opened suction

hole **52h** by adjusting the position of the second closing member **52b**. In other words, the control unit **35** may be capable of controlling the opening degree of the suction inlet **47** by controlling the second adjustment unit **52**.

(66) The control unit **35** may be electrically coupled to the temperature detection unit **53**. The control unit **35** may be capable of controlling the blower **33** and the first adjustment unit **34**, based on the outside air temperature **T** detected by the temperature detection unit **53**.

(67) The control unit **35** may not be electrically coupled to the humidity detection unit **54**. The control unit **35** may be capable of controlling the blower **33** and the first adjustment unit **34**, based on the humidity **H** detected by the humidity detection unit **54**.

(68) The control unit **35** is capable of executing a first mode **M1** and a second mode **M2**. The first mode **M1** and the second mode **M2** are processing relating to heating of the medium **M** supported on the support surface **21a**. Next, the first mode **M1** and the second mode **M2** are described. Here, the step order in each of the modes may be switched in a freely selective manner without departing from the purpose of each of the modes.

(69) First Mode

(70) With reference to FIG. **4**, a routine of the first mode **M1** is described. The first mode **M1** may be executed in a cycle in a state in which power of the printing apparatus **10** is turned on.

(71) As illustrated in FIG. **4**, in Step **S101**, the control unit **35** determines whether the execution condition of the first mode **M1** is satisfied. Here, when printing is being performed, the control unit **35** determines that the execution condition of the first mode **M1** is satisfied. It is determined that the execution condition of the first mode **M1** is not satisfied, which is NO in Step **S101**. The control unit **35** terminates the routine of the first mode **M1**. The control unit **35** determines that the execution condition of the first mode **M1** is satisfied, which is YES in Step **S101**. The control unit **35** proceeds the processing to Step **S102**.

(72) In Step **S102**, the control unit **35** causes the heating section **32** to perform heating. In Step **S103**, the control unit **35** controls the first adjustment unit **34** to adjust the air port **49** to a first opening degree **A1**. The first opening degree **A1** is a set value that is set in advance. In the present exemplary embodiment, the air port **49** adjusted to the first opening degree **A1** is in a fully closed state. In Step **S104**, the control unit **35** controls the second adjustment unit **52** to adjust the suction inlet **47** to a second opening degree **A2**. The second opening degree **A2** is a set value that is set in advance. In the present exemplary embodiment, the suction inlet **47** adjusted to the second opening degree **A2** is in a fully opened state.

(73) In Step **S105**, the control unit **35** controls the blower **33** to adjust an air blowing amount of the blower **33** to a first air blowing amount **W1**. The first air blowing amount **W1** is a set value that is set in advance. In the present exemplary embodiment, when the air blowing amount of the blower **33** is adjusted to the first air blowing amount **W1**, the rotation speed of the fan **33a** of the blower **33** is set to a predetermined rotation speed. The control unit **35** terminates the routine of the first mode **M1**.

(74) The control unit **35** controls the first adjustment unit **34** in the first mode **M1** so as to adjust the amount of air **A** in a state in which heating is performed by the heating section **32**. The control unit **35** controls the second adjustment unit **52** in the first mode **M1** so as to open the suction inlet **47**.

(75) Second Mode

(76) With reference to FIG. **5**, a routine of the second mode **M2** is described. The second mode **M2** may be executed in a cycle in a state in which power of the printing apparatus **10** is turned on.

(77) As illustrated in FIG. **5**, in Step **S201**, the control unit **35** determines whether the execution condition of the second mode **M2** is satisfied. Here, when printing is not being performed, the control unit **35** determines that the execution condition of the second mode **M2** is satisfied. The control unit **35** determines that the execution condition of the second mode **M2** is not satisfied, which is NO in Step **S201**. The control unit **35** terminates the routine of the second mode **M2**. The control unit **35** determines that the execution condition of the second mode **M2** is satisfied, which

is YES in Step S201. The control unit 35 proceeds the processing to Step S202. In Step S202, the control unit 35 stops heating by the heating section 32.

(78) In Step S203, the control unit 35 sets a third opening degree A3, based on the outside air temperature T and the humidity H. The third opening degree A3 is an opening degree larger than the first opening degree A1. In the present exemplary embodiment, the air port 49 adjusted to the third opening degree A3 has an opening degree between a fully closed state and a fully opened state, or is in a fully opened state. When the third opening degree A3 set by the control unit 35 is compared with that under the same condition of the humidity H, there is established such a relationship that the third opening degree A3 is larger as the outside air temperature T is higher. The control unit 35 sets the third opening degree A3 based on the comparison result between the humidity H and the outside air humidity H1. When the third opening degree A3 set by the control unit 35 is compared with that under the same condition of the outside air temperature T, there is established such a relationship that the third opening degree A3 when the humidity H in the region 46 is equal to or higher than the outside air humidity H1 is larger than the third opening degree A3 when the humidity H in the region 46 is lower than the outside air humidity H1. When the humidity H in the region 46 is equal to or higher than the outside air humidity H1, the third opening degree A3 may be a larger opening degree as a value difference between the humidity H in the region 46 and the outside air humidity H1 is greater. When the humidity H in the region 46 is lower than the outside air humidity H1, the third opening degree A3 may be a smaller opening degree as a value difference between the humidity H in the region 46 and the outside air humidity H1 is greater. In Step S204, the control unit 35 controls the first adjustment unit 34 to adjust the air port 49 to the third opening degree A3.

(79) In Step S205, the control unit 35 controls the second adjustment unit 52 to adjust the suction inlet 47 to a fourth opening degree A4. The fourth opening degree A4 is a set value that is set in advance. The fourth opening degree A4 is an opening degree smaller than the second opening degree A2. In the present exemplary embodiment, the suction inlet 47 adjusted to the fourth opening degree A4 has an opening degree between a fully closed state and a fully opened state.

(80) In Step S206, the control unit 35 sets a second air blowing amount W2, based on the outside air temperature T and the humidity H. The second air blowing amount W2 is a value greater than the first air blowing amount W1. In the present exemplary embodiment, when the air blowing amount of the blower 33 is adjusted to the second air blowing amount W2, the rotation speed of the fan 33a of the blower 33 is set higher than the rotation speed that achieves the first air blowing amount W1. In the present exemplary embodiment, when the second air blowing amount W2 set by the control unit 35 is compared with that under the same condition of the humidity H, there is established such a relationship that the second air blowing amount W2 is larger as the outside air temperature T is higher. The control unit 35 sets the second air blowing amount W2 based on the humidity H in the region 46. When the second air blowing amount W2 set by the control unit 35 is compared with that under the same condition of outside air temperature T, there is established such a relationship that the second air blowing amount W2 is larger as the humidity H in the region 46 is higher.

(81) In Step S207, the control unit 35 controls the blower 33 to adjust the air blowing amount of the blower 33 to the second air blowing amount W2. The control unit 35 terminates the routine of the second mode M2.

(82) In Step S202, in the second mode M2, the control unit 35 stops heating by the heating section 32. In Step S204, the control unit 35 adjusts the air port 49 to the third opening degree A3. The third opening degree A3 is an opening degree larger than the first opening degree A1. Thus, the amount of air A obtained through the air port 49 set to the third opening degree A3 is larger than the amount of air A obtained through the air port 49 set to the first opening degree A1. Therefore, it can be said that, in the second mode M2, the control unit 35 controls the first adjustment unit 34 so that the amount of air A is larger than the amount of air A in the first mode M1 in a state in which

heating by the heating section 32 is stopped.

(83) In Step S203, the control unit 35 sets the third opening degree A3 based on the outside air temperature T. When the third opening degree A3 set by the control unit 35 is compared with that under the same condition of the humidity H, there is established such a relationship that the third opening degree A3 is larger as the outside air temperature T is higher. Thus, the amount of air A obtained through the air port 49 set to the third opening degree A3 is larger as the outside air temperature T is higher. Therefore, it can be said that, in the second mode M2, the control unit 35 controls the first adjustment unit 34 so that the amount of air A at a first temperature T1 being the outside air temperature T detected by the temperature detection unit 53 is larger than the amount of air A at a second temperature T2 lower than the first temperature T1.

(84) In Step S203, the control unit 35 sets the third opening degree A3, based on the humidity H. Therefore, it can be said that, in the second mode M2, the control unit 35 controls the first adjustment unit 34, based on the result detected by the humidity detection unit 54.

(85) In Step S206, the control unit 35 sets the second air blowing amount W2, based on the outside air temperature T. In Step S207, the control unit 35 adjusts the air blowing amount of the blower 33 to the second air blowing amount W2. When the second air blowing amount W2 set by the control unit 35 is compared with that under the same condition of the humidity H, there is established such a relationship that the second air blowing amount W2 is larger as the outside air temperature T is higher. Thus, the air flowing toward the blowing outlet 48 by the blower 33 whose blowing amount is set to the second air blowing amount W2 is larger as the outside air temperature T is higher. Therefore, it can be said that, in the second mode M2, the control unit 35 controls the blower 33 so that, when the outside air temperature T detected by the temperature detection unit 53 is the first temperature T1, the air larger than that at the second temperature T2 flows toward the blowing outlet 48.

(86) In Step S206, the control unit 35 sets the second air blowing amount W2, based on the humidity H. Therefore, it can be said that, in the second mode M2, the control unit 35 controls the blower 33, based on the result detected by the humidity detection unit 54.

(87) In Step S205, the control unit 35 adjusts the suction inlet 47 to the fourth opening degree A4. The fourth opening degree A4 is an opening degree smaller than the second opening degree A2. Thus, the amount of the air flowing from the region 46 into the air flow path 43 through the suction inlet 47 set to the fourth opening degree A4 is smaller than the amount of the air obtained through the suction inlet 47 set to the second opening degree A2. Therefore, it can be said that the control unit 35 controls the second adjustment unit 52 so that the opening degree of the suction inlet 47 in the second mode M2 is smaller than the opening degree of the suction inlet 47 in the first mode M1.

(88) Actions of Exemplary Embodiment

(89) Actions of the exemplary embodiment are described.

(90) As illustrated in FIG. 2, in the first mode M1, the heating section 32 performs heating. In the first mode M1, along with heating of the heating section 32, a temperature of the air in the region 46 rises. Along with the rise of the temperature of the air in the region 46, the medium M supported on the support surface 21a is heated. In this manner, the surface of the medium M supported on the support surface 21a is heated, and thus the printing unit 12 is caused to evaporate moisture of the liquid adhering to the medium M. With this, the medium M to which the liquid adheres is dried.

(91) In the first mode M1, driving of the blower 33 causes the air to flow in the air flow path 43. In the first mode M1, the second adjustment unit 52 adjusts the opening degree of the suction inlet 47 so that the suction inlet 47 is in a fully opened state. With this, along with driving of the blower 33, part of the air in the region 46 through the suction inlet 47 into the air flow path 43. The temperature of the air in the region 46 rises due to heating by the heating section 32. Along with driving of the blower 33, the air flowing through the suction inlet 47 into the air flow path 43 flows toward the blowing outlet 48 in the air flow path 43. Part of the air flowing in the air flow path 43

is discharged through the blowing outlet **48** toward the region **46**. Part of the air blown out to the region **46** through the blowing outlet **48** flows downstream in the transport direction D2 along the support surface **21a**. In this manner, the air that is heated by the heating section **32** in the region **46** circulates between the air flow path **43** and the region **46**.

(92) In particular, in the first mode M1, the first adjustment unit **34** adjusts the opening degree of the air port **49** so that the air port **49** is in a fully closed state. Thus, the air flowing through the air port **49** into the air flow path **43** is suppressed. The air flowing through the air port **49** into the air flow path **43** has a temperature lower than the temperature of the air in the region **46** between the heating unit **25** and the support surface **21a**. Therefore, the air flowing through the air port **49** into the air flow path **43** is suppressed, and thus temperature fall of the air in the air flow path **43** can be suppressed. Along with this, temperature fall of the air blowing out through the blowing outlet **48** toward the support surface **21a** can be suppressed.

(93) Meanwhile, in the second mode M2, heating by the heating section **32** is stopped which is different from the first mode M1. Along with this, the temperature of the air in the region **46** does not rise, and hence the temperature of the air in the region **46** is lower in the second mode M2, as compared to the temperature in the first mode M1. Therefore, excessive heating by the heating section **32** with respect to the medium M transported along the support surface **21a** can be suppressed. Thus, a thermal damage caused to the medium M supported on the support surface **21a** can be suppressed.

(94) Further, in the second mode M2, driving of the blower **33** also causes the air to flow in the air flow path **43** similarly to the first mode M1. In the second mode M2, the second adjustment unit **52** adjusts the opening degree of the suction inlet **47** so that the opening degree of the suction inlet **47** is smaller than the opening degree of the suction inlet **47** in the first mode M1. With this, the amount of the air flowing from the region **46** into the air flow path **43** through the suction inlet **47** along with driving of the blower **33** is smaller than that in the first mode M1. In this manner, the air that is heated by the heating section **32** in the region **46** is less likely to circulate between the air flow path **43** and the region **46**.

(95) Further, in the second mode M2, the first adjustment unit **34** adjusts the opening degree of the air port **49** so as to open the air port **49**. With this, along with driving of the blower **33**, part of the outside air flows through the air port **49** into the air flow path **43**. Part of the outside air flowing through the air port **49** into the air flow path **43** along with driving of the blower **33** flows toward the blowing outlet **48** in the air flow path **43**. Part of the air flowing in the air flow path **43** is discharged through the blowing outlet **48** toward the region **46**. Part of the air blown out to the region **46** through the blowing outlet **48** flows downstream in the transport direction D2 along the support surface **21a**. In this manner, the outside air flowing through the air port **49** into the air flow path **43** is blown to the region **46**. The temperature of the outside air flowing through the air port **49** into the air flow path **43** is lower than the temperature in the air in the region **46**. Therefore, the outside air flowing through the air port **49** into the air flow path **43** is discharged through the blowing outlet **48** to the region **46**, and thus temperature fall of the air flow path **43** can be promoted. Along with this, temperature fall of the air blowing out through the blowing outlet **48** toward the support surface **21a** can be promoted. Further, the air that is heated by the heating section **32** in the region **46** is discharged downstream in the transport direction D2 from the region **46**.

(96) Effects of Exemplary Embodiment

(97) Effects of the exemplary embodiment are described.

(98) (1) The first adjustment unit **34** is capable of adjusting the amount of air A flowing into the air flow path **43** from the outside of the duct **31** through the air port **49**. The air port **49** is provided in the non-facing region **45** of the duct **31**, which does not face the support surface **21a**. Thus, the air having a temperature lower than the air in the region **46** between the heating unit **25** and the support surface **21a** can flow through the air port **49** into the air flow path **43**. In the first mode M1,

heating by the heating section 32 is performed, and the first adjustment unit 34 adjusts the amount of air A. In the second mode M2, heating by the heating section 32 is stopped, and the first adjustment unit 34 performs adjustment so that the amount of air A in the second mode M2 is larger than the amount of air A in the first mode M1. Thus, in the second mode M2, the temperature of the air that flows in the air flow path 43 and is discharged through the blowing outlet 48 toward the support surface 21a can be lowered than that in the first mode M1. In the second mode M2, the temperature of the air in the region 46 between the heating unit 25 and the support surface 21a is easily lowered. Therefore, a thermal damage caused to the medium M transported along the support surface 21a can be suppressed.

(99) (2) In the second mode M2, the control unit 35 controls the first adjustment unit 34 so that the amount of air A at a first temperature T1 being the outside air temperature T detected by the temperature detection unit 53 is larger than the amount of air A at a second temperature T2 lower than the first temperature T1. In the second mode M2, the air flowing through the air port 49 into the air flow path 43 at the outside air temperature T being the first temperature T1 is larger than that at the second temperature T2 lower than the first temperature T1. Thus, even when the temperature of the air flowing through the air port 49 into the air flow path 43 is relatively high, the air that flows in the air flow path 43 and is discharged through the blowing outlet 48 toward the support surface 21a is increased. With this, the temperature of the air in the region 46 between the heating unit 25 and the support surface 21a is easily lowered. Therefore, a thermal damage caused to the medium M transported along the support surface 21a can suitably be suppressed in accordance with the outside air temperature T.

(100) (3) In the second mode M2, when the outside air temperature T detected by the temperature detection unit 53 is the first temperature T1, the control unit 35 controls the blower 33 so that the air larger than that in the second temperature T2 flows toward the blowing outlet 48. In the second mode M2, the amount of the air in the air flow path 43, which flows toward the blowing outlet 48, at the outside air temperature T being the first temperature T1 is larger than that at the second temperature T2 lower than the first temperature T1. Thus, even when the temperature of the air flowing through the air port 49 into the air flow path 43 is relatively high, the air that flows in the air flow path 43 and is discharged through the blowing outlet 48 toward the support surface 21a is increased. With this, the temperature of the air in the region 46 between the heating unit 25 and the support surface 21a is easily lowered. Therefore, a thermal damage caused to the medium M transported along the support surface 21a can suitably be suppressed to a further extent in accordance with the outside air temperature T.

(101) (4) The heating unit 25 includes the humidity detection unit 54 that detects the humidity H in the region 46 between the heating unit 25 and the support surface 21a. In the second mode M2, the control unit 35 controls the blower 33 and the first adjustment unit 34, based on the result detected by the humidity detection unit 54. Therefore, in the second mode M2, the blower 33 and the first adjustment unit 34 can be controlled based on the humidity H in the region 46 between the heating unit 25 and the support surface 21a.

(102) (5) The second adjustment unit 52 is capable of adjusting the opening degree of the suction inlet 47. As the opening degree of the suction inlet 47 adjusted by the second adjustment unit 52 is larger, the air flowing from the region 46 between the heating unit 25 and the support surface 21a through the suction inlet 47 into the air flow path 43 is increased. In the second mode M2, the second adjustment unit 52 performs adjustment so that the opening degree of the suction inlet 47 is smaller than the opening degree of the suction inlet 47 in the first mode M1. Thus, the opening degree of the suction inlet 47 in the first mode M1 is adjusted to an opening degree larger than that in the second mode M2, and hence the warm air in the region 46 between the heating unit 25 and the support surface 21a easily flows through the suction inlet 47 into the air flow path 43.

Therefore, the temperature of the air flowing in the air flow path 43 can be raised. Thus, the air discharged through the blowing outlet 48 from the air flow path 43 toward the support surface 21a

easily raises the temperature of the air in the region **46** between the heating unit **25** and the support surface **21a**. Therefore, evaporation of moisture of the liquid adhering to the medium **M** transported along the support surface **21a** and drying of the medium **M** to which the liquid adheres can further be promoted. The suction inlet **47** in the second mode **M2** is adjusted to have an opening degree smaller than the opening degree of the suction inlet **47** in the first mode **M1**. Thus, the warm air in the region **46** between the heating unit **25** and the support surface **21a** is less likely to flow through the suction inlet **47** into the air flow path **43**. Therefore, rise of the temperature of the air flowing in the air flow path **43** can be suppressed. Thus, the air discharged through the blowing outlet **48** from the air flow path **43** toward the support surface **21a** easily lowers the temperature of the air in the region **46** between the heating unit **25** and the support surface **21a**. Therefore, a thermal damage caused to the medium **M** transported along the support surface **21a** can further be suppressed.

(103) (6) The air port **49** is in communication with the air flow path downstream of the parallel installation portion **43a** being a portion of the air flow path **43**, which is aligned with the heating section **32** through intermediation of the partition wall **42** in the orthogonal direction **D3** orthogonal to the support surface **21a**. Thus, the air flowing through the air port **49** from the outside of the duct **31** into the air flow path **43** is less likely to flow through the parallel installation portion **43a**. Therefore, the cooler air can be discharged through the blowing outlet **48** from the air flow path **43** toward the support surface **21a**.

(104) (7) The partition wall **42** that separates the air flow path **43** and the heating section **32** contains the heat insulating material **42a**. Thus, the air flowing through the air port **49** from the outside of the duct **31** into the air flow path **43** is less likely to be heated by the heating section **32**. Therefore, the cooler air can be discharged through the blowing outlet **48** from the air flow path **43** toward the support surface **21a**.

(105) (8) The drying device **20** includes the removal member **51** that removes a foreign matter from the air flowing through the air port **49**. Thus, a foreign matter can be prevented from entering the air flow path **43** through the air port **49**.

(106) (9) In the second mode **M2**, when the outside air temperature **T** detected by the temperature detection unit **53** is the first temperature **T1**, the control unit **35** controls the first adjustment unit **34** so that the amount of air **A** is larger than that at the second temperature **T2** lower than the first temperature **T1**. In the second mode **M2**, the air flowing through the air port **49** into the air flow path **43** at the outside air temperature **T** being the second temperature **T2** is smaller than that at the first temperature **T1** higher than the second temperature **T2**. Thus, when the outside air temperature **T** is the second temperature **T2**, the temperature of the air flowing in the air flow path **43** is less likely to be lowered as compared to a case in which the outside air temperature **T** is the first temperature **T1**. Therefore, in a state in which the outside air temperature **T** is low enough to satisfactorily suppress a thermal damage caused to the medium **M** with a little amount of air **A**, the first adjustment unit **34** adjusts the amount of air **A** to be smaller. With this, the duct **31** can be maintained under a heated state. A subsequent first mode **M1** can be executed in a state in which the duct **31** is warmed. Thus, the temperature of the air blown out from the air flow path **43** through the suction inlet **47** onto the support surface **21a** can be raised rapidly.

(107) Modified Examples

(108) The exemplary embodiment described above may be modified as follows. The exemplary embodiment and modified examples thereof to be described below may be implemented in combination within a range in which a technical contradiction does not arise.

(109) The printing apparatus **10** may include an encoder that detects a driving amount of the transport motor. In other words, the drying device **20** may include an encoder. The control unit **35** may acquire a transport position of the medium **M** from a detection value of the encoder. The control unit **35** may determine, based on the acquired transport position of the medium **M**, whether transport of the medium **M** is started before a predetermined time period longer than a stoppage time of intermittent transport elapses. The execution condition of the first mode **M1** may include

that transport of the medium M is started before the predetermined time period elapses. When printing is being performed, and transport of the medium M is started before the predetermined time period elapses, the control unit **35** may determine that the execution condition of the first mode **M1** is satisfied. The execution condition of the second mode **M2** may include that transport of the medium M is not started before the predetermined time period elapses. When printing is being performed, and transport of the medium M is not started before the predetermined time period elapses, the control unit **35** may determine that the execution condition of the second mode **M2** is satisfied.

(110) The position at which the removal member **51** is provided is not limited to the inside of the air port **49**. For example, the removal member **51** may be provided to the outside of the duct **31** so as to close the opening of the air port **49** oriented to the outside of the duct **31**. The removal member **51** may be provided to the inside of the duct **31** so as to close the opening of the air port **49** oriented to the inside of the duct **31**.

(111) The removal member **51** may be omitted from the drying device **20**.

(112) The heat insulating material **42a** may be away from the inner wall **42b**. The heat insulating material **42a** may be positioned between the inner wall **42b** and the second outer wall **41b**.

(113) The entire partition wall **42** may be the heat insulating material **42a**. In this case, the partition wall **42** also contains the heat insulating material **42a**.

(114) The heat insulating material **42a** may be omitted from the partition wall **42**. In this case, for example, the entire partition wall **42** may be the inner wall **42b**. In this case, along with heating by the heating section **32**, the heat easily propagates to the air in the air flow path **43** via the partition wall **42**. Thus, the heating section **32** heats the air in the air flow path **43**.

(115) The air port **49** may communicate with the parallel installation portion **43a** or communicate with upstream of the parallel installation portion **43a** in the air flow path **43**.

(116) The communication position of the air port **49** with the air flow path **43** is not limited to a position between the suction inlet **47** and the blowing outlet **48**.

(117) The suction inlet **47** may be provided in the non-facing region **45** as long as the air in the region **46** can flow through the suction inlet **47** into the air flow path **43**.

(118) The blowing outlet **48** may be provided in the non-facing region **45** as long as the air can be blown out toward the support surface **21a**.

(119) The partition wall **42** may be omitted from the heating unit **25**. In this case, for example, the space that is not partitioned by the partition wall **42** is formed inside the duct **31**. The heating section **32** and the air flow path **43** may be positioned in the space inside the duct **31**.

(120) The control unit **35** may control the second adjustment unit **52** so that the suction inlet **47** is not opened in the second mode **M2**. When the suction inlet **47** in this case is adjusted to the fourth opening degree **A4**, the suction inlet **47** is in a fully closed state. In the second mode **M2**, the air in the region **46** does not flow through the suction inlet **47** into the air flow path **43**.

(121) The installation position of the second adjustment unit **52** is not limited to the air flow path **43**. For example, the second adjustment unit **52** may be positioned outside of the duct **31**. The second adjustment unit **52** includes the second main body portion **52a** in which the suction hole **52h** is formed and the second closing member **52b** capable of closing the suction hole **52h**, but is not limited thereto. For example, the second adjustment unit **52** may include an opening/closing valve capable of opening and closing the suction inlet **47**. With this displacement of the opening/closing valve, the suction inlet **47** may be adjusted to be in a fully closed state or a fully opened state, or have an opening degree between a fully closed state and a fully opened state.

(122) The second adjustment unit **52** may be omitted from the heating unit **25**. When any one of the first mode **M1** and the second mode **M2** is executed, the suction inlet **47** in this case is opened in a fully opened state. Step **S104** may be omitted from the routine illustrated in FIG. **4**. Step **S205** may be omitted from the routine illustrated in FIG. **5**.

(123) In the second mode **M2**, the control unit **35** may control only one of the blower **33** and the

first adjustment unit **34**, based on the result detected by the humidity detection unit **54**. In the second mode **M2**, when the control unit **35** controls only the blower **33**, based on the result detected by the humidity detection unit **54**, the processing in Step **S203** in the routine illustrated in FIG. **5** may be changed to the processing for setting the third opening degree **A3** based on the outside air temperature **T**. In the second mode **M2**, when the control unit **35** controls only the first adjustment unit **34**, based on the result detected by the humidity detection unit **54**, the processing in Step **S206** in the routine illustrated in FIG. **5** may be changed to the processing for setting the second air blowing amount **W2** based on the outside air temperature **T**.

(124) In the second mode **M2**, the control unit **35** may not perform control based on the result detected by the humidity detection unit **54**. When the control unit **35** does not perform control based on the result detected by the humidity detection unit **54**, the humidity detection unit **54** may be omitted from the heating unit **25**.

(125) In the second mode **M2**, the control unit **35** may select one air blowing amount from a plurality of air blowing amounts that are different from each other in a stepwise manner, in accordance with the outside air temperature **T**, set the selected air blowing amount as the second air blowing amount **W2**, and thus adjust the blower **33**. Specifically, the air blowing amount at the outside air temperature **T** within a temperature range including the second temperature **T2** is referred to as a low-temperature air blowing amount. The air blowing amount at the outside air temperature **T** within a temperature range including the first temperature **T1** higher than the second temperature **T2** is referred to as a high-temperature air blowing amount. The high-temperature air blowing amount is an air blowing amount larger than the low-temperature air blowing amount. When the outside air temperature **T** is within the temperature range corresponding to the high-temperature air blowing amount, the high-temperature air blowing amount is adopted as the second air blowing amount **W2**. When the outside air temperature **T** is within the temperature range corresponding to the low-temperature air blowing amount, the low-temperature air blowing amount is adopted as the second air blowing amount **W2**. In this case, in the second mode **M2**, it can also be said that the control unit **35** controls the blower **33** so that, when the outside air temperature **T** detected by the temperature detection unit **53** is the first temperature **T1**, the air larger than that at the second temperature **T2** flows toward the blowing outlet **48**.

(126) The control unit **35** may not control the blower **33**, based on the result detected by the temperature detection unit **53**. In this case, the processing in Step **S206** in the routine illustrated in FIG. **5** may be changed to the processing for setting the second air blowing amount **W2** based on the humidity **H**.

(127) In the second mode **M2**, the control unit **35** may select one opening degree from a plurality of opening degrees that are different from each other in a stepwise manner, in accordance with the outside air temperature **T**, set the selected opening degree as the third opening degree **A3**, and thus adjust the air port **49**. Specifically, the opening degree at the outside air temperature **T** within the temperature range including the second temperature **T2** is referred to as a low-temperature opening degree. The opening degree at the outside air temperature **T** within the temperature range including the first temperature **T1** higher than the second temperature **T2** is referred to as a high-temperature opening degree. The low-temperature opening degree is an opening degree smaller than the high-temperature opening degree. When the outside air temperature **T** is within the temperature range corresponding to the high-temperature opening degree, the high-temperature opening degree is adopted as the third opening degree **A3**. When the outside air temperature **T** is within the temperature range corresponding to the low-temperature opening degree, the low-temperature opening degree is adopted as the third opening degree. In this case, in the second mode **M2**, it can also be said that the control unit **35** controls the first adjustment unit **34** so that, when the outside air temperature **T** detected by the temperature detection unit **53** is the first temperature **T1**, the amount of air **A** is larger than the amount of air **A** at the second temperature **T2** lower than the first temperature **T1**.

(128) The control unit **35** may not control the first adjustment unit **34**, based on the result detected by the temperature detection unit **53**. In this case, the processing in Step **S203** in the routine illustrated in FIG. **5** may be changed to the processing for setting the third opening degree **A3** based on the humidity **H**.

(129) In the second mode **M2**, the control unit **35** may control the blower **33** through use of the second air blowing amount **W2** being a set value that is set in advance. The second air blowing amount **W2** in this case may be a value greater than the first air blowing amount **W1**, a value equal to the first air blowing amount **W1**, or a value less than the first air blowing amount **W1**. In this case, the processing in Step **S206** may be omitted from the routine illustrated in FIG. **5**.

(130) In the second mode **M2**, the control unit **35** may control the first adjustment unit **34** through use of the third opening degree **A3** being a set value that is set in advance. In this case, the processing in Step **S203** may be omitted from the routine illustrated in FIG. **5**. The third opening degree **A3** in this case is an opening degree larger than the first opening degree **A1**. Therefore, in this case, it can also be said that, in the second mode **M2**, the first adjustment unit **34** is controlled so that the amount of air **A** in the second mode **M2** is larger than the amount of air **A** in the first mode **M1**.

(131) In the second mode **M2**, the control unit **35** may not perform control based on the result detected by the temperature detection unit **53**. When the control unit **35** does not perform control based on the result detected by the temperature detection unit **53**, the temperature detection unit **53** may be omitted from the heating unit **25**.

(132) The installation position of the first adjustment unit **34** is not limited to the air flow path **43**. For example, the first adjustment unit **34** may be positioned outside of the duct **31**. The first adjustment unit **34** includes the first main body portion **34a** in which the air hole **34h** is formed and the first closing member **34b** capable of closing the air hole **34h**, but is not limited thereto. For example, the first adjustment unit **34** may include an opening/closing valve capable of opening and closing the air port **49**. With this displacement of the opening/closing valve, the air port **49** may be adjusted to be in a fully closed state or a fully opened state, or have an opening degree between a fully closed state and a fully opened state.

(133) The printing apparatus **10** may be a liquid ejection device that jets or ejects a liquid other than ink. A state of the liquid ejected as a fine liquid droplet from the liquid ejection device includes a granular state, a tear drop state, a string-like shape, and the like. The liquid described herein is only required to be a material that can be ejected from the liquid ejection device. For example, the liquid may only be required to have a state in which a substance is in a liquid phase, and includes fluid such as liquid with high or low viscosity, sol, gel water, or other inorganic solvents, an organic solvent, a solution, a liquid resin, a liquid metal, and a metallic melt. The liquid includes not only a liquid being a state of a substance, but also a liquid in which particles of functional materials formed of solid materials such as pigments and metallic particles are dissolved, dispersed, or mixed in a solvent, or the like. Typical examples of the liquid include ink as described in the above-mentioned exemplary embodiment, a liquid crystal, and the like. Herein, the ink includes general aqueous ink, solvent ink, and various types of liquid compositions such as gel ink and hot-melt ink. For example, specific examples of the liquid ejection device include a device that ejects ink including materials such as an electrode material and a color material used in manufacture of liquid crystal displays, electroluminescent displays, surface light emitting displays, color filters and the like in a dispersed or dissolved form. Additionally, the liquid ejection device may be a device that ejects bioorganic substances used for biochip manufacturing, a device that is used as a precision pipette and ejects a liquid to be a sample, a printing apparatus, a micro dispenser, or the like. The liquid ejection device may be a device that ejects lubricant to a precision machine such as a clock or a camera in a pinpoint manner, and a device that ejects a transparent resin liquid such as an ultraviolet cure resin or the like on a substrate for forming a tiny hemispherical lens, an optical lens, or the like used for an optical communication element and the like. The liquid ejection device

may be a device that ejects an acid or alkaline etching solution for subjecting a substrate or the like to etching.

(134) Supplementary Note

(135) Hereinafter, technical concepts and effects thereof that are understood from the above-described embodiment and modified examples are described.

(136) (A) A drying device includes a support unit configured to support, on a support surface, a medium transported after printing, a heating unit facing the support surface and being configured to heat the medium supported on the support surface after printing, and a control unit configured to control the heating unit. In the drying device, the heating unit includes a duct having a blowing outlet through which air is blown out toward the support surface, a suction inlet through which the air blown out through the blowing outlet toward the support surface is sucked, and an air port being different from the blowing outlet and the suction inlet, and including an air flow path through which the air is flowable, a heating section configured to heat at least one of the air in the air flow path and the medium supported on the support surface, a blower configured to cause the air in the air flow path to flow toward the blowing outlet, and a first adjustment unit configured to adjust an amount of air flowing into the air flow path from an outside of the duct through the air port, the duct has a facing region facing the support surface and a non-facing region not facing the support surface, the air port is provided in the non-facing region, and the control unit is configured to execute a first mode in which the first adjustment unit is controlled to adjust the amount of air in a state in which the heating section performs heating and a second mode in which the first adjustment unit is controlled to cause the amount of air to be more than the amount of air in the first mode in a state in which heating by the heating section is stopped.

(137) With this configuration, the first adjustment unit is capable of adjusting the amount of air flowing into the air flow path from the outside of the duct through the air port. The air port is provided in the non-facing region of the duct, which does not face the support surface. Thus, the air having a temperature lower than the air in the region between the heating unit and the support surface can flow through the air port into the air flow path. In the first mode, heating by the heating section is performed, and the first adjustment unit adjusts the amount of air. In the second mode, heating by the heating section is stopped, and the first adjustment unit performs adjustment so that the amount of air in the second mode is larger than the amount of air in the first mode. Thus, in the second mode, the temperature of the air that flows in the air flow path and is discharged through the blowing outlet toward the support surface can be lowered than that in the first mode, and hence, in the second mode, the temperature of the air in the region between the heating unit and the support surface is easily lowered. Therefore, a thermal damage caused to the medium transported along the support surface can be suppressed.

(138) (B) In the drying device, the heating unit includes a temperature detection unit configured to detect an outside air temperature, and, in the second mode, the control unit controls the first adjustment unit to cause the amount of air when the outside air temperature detected by the temperature detection unit is a first temperature to be more the amount of air when the detected outside air temperature is a second temperature lower than the first temperature.

(139) With this configuration, in the second mode, the air flowing through the air port into the air flow path at the outside air temperature being the first temperature is larger than that at the second temperature lower than the first temperature. Thus, even when the temperature of the air flowing through the air port into the air flow path is relatively high, the air that flows in the air flow path and is discharged through the blowing outlet toward the support surface is increased, and hence the temperature of the air in the region between the heating unit and the support surface is easily lowered. Therefore, a thermal damage caused to the medium transported along the support surface can suitably be suppressed in accordance with the outside air temperature.

(140) (C) In the drying device, in the second mode, the control unit controls the blower to cause air to flow toward the blowing outlet when the outside air temperature detected by the temperature

detection unit is the first temperature, the air being more than that at the second temperature.

(141) With this configuration, in the second mode, the amount of the air in the air flow path, which flows toward the blowing outlet, at the outside air temperature being the first temperature is larger than that at the second temperature lower than the first temperature. Thus, even when the temperature of the air flowing through the air port into the air flow path is relatively high, the air that flows in the air flow path and is discharged through the blowing outlet toward the support surface is increased, and hence the temperature of the air in the region between the heating unit and the support surface is easily lowered. Therefore, a thermal damage caused to the medium transported along the support surface can suitably be suppressed to a further extent in accordance with the outside air temperature.

(142) (D) In the drying device, the heating unit includes a humidity detection unit configured to detect humidity in a region between the heating unit and the support surface, and, in the second mode, the control unit controls the blower and the first adjustment unit, based on a result detected by the humidity detection unit.

(143) With this configuration, in the second mode, the blower and the first adjustment unit can be controlled based on the humidity in the region between the heating unit and the support surface.

(144) (E) In the drying device, the heating unit includes a second adjustment unit configured to adjust an opening degree of the suction inlet, and the control unit controls the second adjustment unit to open the suction inlet in the first mode, and controls the second adjustment unit to cause an opening degree of the suction inlet in the second mode to be smaller than an opening degree of the suction inlet in the first mode.

(145) With this configuration, second adjustment unit is capable of adjusting the opening degree of the suction inlet. As the opening degree of the suction inlet adjusted by the second adjustment unit is larger, the air flowing from the region between the heating unit and the support surface through the suction inlet into the air flow path is increased. In the second mode, the second adjustment unit performs adjustment so that the opening degree of the suction inlet is smaller than the opening degree of the suction inlet in the first mode. Here, the opening degree of the suction port, which is smaller than the opening degree of the suction port in the first mode, includes an opening degree at which the suction port is not opened. Thus, the opening degree of the suction inlet in the first mode is adjusted to an opening degree larger than that in the second mode, and hence the warm air in the region between the heating unit and the support surface easily flows through the suction inlet into the air flow path. Therefore, the temperature of the air flowing in the air flow path can be raised. Thus, the air discharged through the blowing outlet from the air flow path toward the support surface easily raises the temperature of the air in the region between the heating unit and the support surface. Therefore, evaporation of moisture of the liquid adhering to the medium transported along the support surface and drying of the medium to which the liquid adheres can further be promoted. The suction port in the second mode is adjusted to have an opening degree smaller than the opening degree of the suction port in the first mode. Thus, the warm air in the region between the heating unit and the support surface is less likely to flow through the suction inlet into the air flow path. Therefore, rise of the temperature of the air flowing in the air flow path can be suppressed. Thus, the air discharged through the blowing outlet from the air flow path toward the support surface easily lowers the temperature of the air in the region between the heating unit and the support surface. Therefore, a thermal damage caused to the medium transported along the support surface can further be suppressed.

(146) (F) In the drying device, the heating unit has, inside the duct, a partition wall that separates the air flow path and the heating section, and when, in the air flow path, a portion thereof aligned with the heating section through intermediation of the partition wall in an orthogonal direction orthogonal to the support surface is regarded as a parallel installation portion, the air port is in communication with the air flow path downstream of the parallel installation portion in a flowing direction of air flowing through the air flow path.

(147) With this configuration, the air port is in communication with the air flow path downstream of the parallel installation portion being a portion of the air flow path, which is aligned with the heating section through intermediation of the partition wall in the orthogonal direction orthogonal to the support surface. Thus, the air flowing through the air port from the outside of the duct into the air flow path is less likely to flow through the parallel installation portion. Therefore, the cooler air can be discharged through the blowing outlet from the air flow path toward the support surface.

(148) (G) In the drying device, the partition wall contains a heat insulating material.

(149) With this configuration, the partition wall that separates the air flow path and the heating section contains the heat insulating material. Thus, the air flowing through the air port from the outside of the duct into the air flow path is less likely to be heated by the heating section. Therefore, the cooler air can be discharged through the blowing outlet from the air flow path toward the support surface.

(150) (H) The drying device further includes a removal member configured to remove a foreign matter from the air flowing through the air port.

(151) With this configuration, the drying device includes the removal member that removes a foreign matter from the air flowing through the air port. Thus, a foreign matter can be prevented from entering the air flow path through the air port.

(152) (I) A printing apparatus includes a transport unit configured to transport a medium, a printing unit configured to perform printing on the medium transported by the transport unit, a support unit configured to support, on a support surface, the medium after printing performed by the printing unit, a heating unit facing the support surface and being configured to heat the medium supported on the support surface after printing, and a control unit configured to control the heating unit. In the printing apparatus, the heating unit includes a duct having a blowing outlet through which air is blown out toward the support surface, a suction inlet through which the air blown out through the blowing outlet toward the support surface is sucked, and an air port being different from the blowing outlet and the suction inlet, and including an air flow path through which the air is flowable, a heating section configured to heat at least one of the air in the air flow path and the medium supported on the support surface, a blower configured to cause the air in the air flow path to flow toward the blowing outlet, and a first adjustment unit configured to adjust an amount of air flowing into the air flow path from an outside of the duct through the air port, the duct has a facing region facing the support surface and a non-facing region not facing the support surface, the air port is provided in the non-facing region, and the control unit is configured to execute a first mode in which the first adjustment unit is controlled to adjust the amount of air in a state in which the heating section performs heating and a second mode in which the first adjustment unit is controlled to cause the amount of air to be more than the amount of air in the first mode in a state in which heating by the heating section is stopped.

(153) With this configuration, the first adjustment unit is capable of adjusting the amount of air flowing into the air flow path from the outside of the duct through the air port. The air port is provided in the non-facing region of the duct, which does not face the support surface. Thus, the air having a temperature lower than the air in the region between the heating unit and the support surface can flow through the air port into the air flow path. In the first mode, heating by the heating section is performed, and the first adjustment unit adjusts the amount of air. In the second mode, heating by the heating section is stopped, and the first adjustment unit performs adjustment so that the amount of air in the second mode is larger than the amount of air in the first mode. Thus, in the second mode, the temperature of the air that flows in the air flow path and is discharged through the blowing outlet toward the support surface can be lowered than that in the first mode, and hence, in the second mode, the temperature of the air in the region between the heating unit and the support surface is easily lowered. Therefore, a thermal damage caused to the medium transported along the support surface can be suppressed.

Claims

1. A drying device, comprising: a support unit configured to support, on a support surface, a medium transported after printing; a heating unit facing the support surface and being configured to heat the medium supported on the support surface after printing; and a control unit configured to control the heating unit, wherein the heating unit includes: a duct having a blowing outlet through which air is blown out toward the support surface, a suction inlet through which the air blown out through the blowing outlet toward the support surface is sucked, and an air port being different from the blowing outlet and the suction inlet, and including an air flow path through which the air is flowable; a heating section configured to heat at least one of the air in the air flow path and the medium supported on the support surface; a blower configured to cause the air in the air flow path to flow toward the blowing outlet; and a first adjustment unit configured to adjust an amount of air flowing into the air flow path from an outside of the duct through the air port, the duct has a facing region facing the support surface and a non-facing region not facing the support surface, the air port is disposed in the non-facing region, and the control unit is configured to execute a first mode in which the first adjustment unit is controlled to adjust the amount of air in a state in which the heating section performs heating and a second mode in which the first adjustment unit is controlled to cause the amount of air to be more than the amount of air in the first mode in a state in which heating by the heating section is stopped, wherein the heating unit includes a second adjustment unit configured to adjust an opening degree of the suction inlet, and the control unit controls the second adjustment unit to open the suction inlet in the first mode, and controls the second adjustment unit to cause an opening degree of the suction inlet in the second mode to be smaller than an opening degree of the suction inlet in the first mode.
2. The drying device according to claim 1, wherein the heating unit includes a humidity detection unit configured to detect humidity in a region between the heating unit and the support surface, and in the second mode, the control unit controls the blower and the first adjustment unit, based on a result detected by the humidity detection unit.
3. The drying device according to claim 1, wherein the heating unit has, inside the duct, a partition wall that separates the air flow path and the heating section, and when, in the air flow path, a portion thereof aligned with the heating section through intermediation of the partition wall in an orthogonal direction orthogonal to the support surface is regarded as a parallel installation portion, the air port is in communication with the air flow path downstream of the parallel installation portion in a flowing direction of air flowing through the air flow path.
4. The drying device according to claim 3, wherein the partition wall contains a heat insulating material.
5. The drying device according to claim 1, comprising a removal member configured to remove a foreign matter from the air flowing through the air port.
6. A drying device, comprising: a support unit configured to support, on a support surface, a medium transported after printing; a heating unit facing the support surface and being configured to heat the medium supported on the support surface after printing; and a control unit configured to control the heating unit, wherein the heating unit includes: a duct having a blowing outlet through which air is blown out toward the support surface, a suction inlet through which the air blown out through the blowing outlet toward the support surface is sucked, and an air port being different from the blowing outlet and the suction inlet, and including an air flow path through which the air is flowable; a heating section configured to heat at least one of the air in the air flow path and the medium supported on the support surface; a blower configured to cause the air in the air flow path to flow toward the blowing outlet; and a first adjustment unit configured to adjust an amount of air flowing into the air flow path from an outside of the duct through the air port, the duct has a facing region facing the support surface and a non-facing region not facing the support surface, the air

port is disposed in the non-facing region, and the control unit is configured to execute a first mode in which the first adjustment unit is controlled to adjust the amount of air in a state in which the heating section performs heating and a second mode in which the first adjustment unit is controlled to cause the amount of air to be more than the amount of air in the first mode in a state in which heating by the heating section is stopped, wherein the heating unit includes a temperature detection unit configured to detect an outside air temperature, and in the second mode, the control unit controls the first adjustment unit to increase the amount of air when the outside air temperature detected by the temperature detection unit is a first temperature to be more than the amount of air when the detected outside air temperature is a second temperature lower than the first temperature.

7. The drying device according to claim 6, wherein in the second mode, the control unit controls the blower to cause more air to flow toward the blowing outlet when the outside air temperature detected by the temperature detection unit is the first temperature than when the detected outside air temperature is the second temperature.

8. A printing apparatus, comprising: a transport unit configured to transport a medium; a printing unit configured to perform printing on the medium transported by the transport unit; a support unit configured to support, on a support surface, the medium after printing performed by the printing unit; a heating unit facing the support surface and being configured to heat the medium supported on the support surface after printing; and a control unit configured to control the heating unit, wherein the heating unit includes: a duct having a blowing outlet through which air is blown out toward the support surface, a suction inlet through which the air blown out through the blowing outlet toward the support surface is sucked, and an air port being different from the blowing outlet and the suction inlet, and including an air flow path through which the air is flowable; a heating section configured to heat at least one of the air in the air flow path and the medium supported on the support surface; a blower configured to cause the air in the air flow path to flow toward the blowing outlet; and a first adjustment unit configured to adjust an amount of air flowing into the air flow path from an outside of the duct through the air port, the duct has a facing region facing the support surface and a non-facing region not facing the support surface, the air port is disposed in the non-facing region, and the control unit is configured to execute a first mode in which the first adjustment unit is controlled to adjust the amount of air in a state in which the heating section performs heating and a second mode in which the first adjustment unit is controlled to cause the amount of air to be more than the amount of air in the first mode in a state in which heating by the heating section is stopped, wherein the heating unit includes a second adjustment unit configured to adjust an opening degree of the suction inlet, and the control unit controls the second adjustment unit to open the suction inlet in the first mode, and controls the second adjustment unit to cause an opening degree of the suction inlet in the second mode to be smaller than an opening degree of the suction inlet in the first mode.
